

ASCII MASTER FLOW – PANEL 1/12: Bit and qubit card

CLASSICAL BIT -> recorded as 0 or 1

QUBIT -> amplitudes over basis states

MEASUREMENT -> one classical outcome with probability

SPEED -> not the defining difference

RULE -> state description is not two stored classical bits

ASCII MASTER FLOW – PANEL 2/12: Quantum principle split

SUPERPOSITION -> one system in a combination of basis states

ENTANGLEMENT -> joint state of two or more systems

NO-CLONING -> arbitrary unknown state cannot be copied perfectly

NO SIGNALLING -> entanglement does not transmit messages faster

RULE -> each principle answers a different question

ASCII MASTER FLOW – PANEL 3/12: Algorithm mechanism rail

PREPARE STATE -> initialise qubits

QUANTUM GATES -> transform amplitudes

INTERFERENCE -> suppress and reinforce paths

MEASUREMENT -> obtain classical sample

TRAP -> superposition does not reveal every answer

ASCII MASTER FLOW – PANEL 4/12: Decoherence engineering stack

ENVIRONMENTAL INTERACTION -> phase information degrades
COHERENCE TIME -> finite computation window
CRYOGENICS / VACUUM / SHIELDING -> reduce disturbance
CONTROL ELECTRONICS / MATERIALS -> execute accurate gates
OUTCOME -> engineering, not theory alone, limits scale

ASCII MASTER FLOW – PANEL 5/12: Physical-to-logical ladder

PHYSICAL QUBIT -> noisy hardware element
ERROR CHARACTERISATION -> measure failure channels
ERROR-CORRECTING CODE -> redundant encoding
LOGICAL QUBIT -> protected information unit
RULE -> raw qubit count is not useful capability

ASCII MASTER FLOW – PANEL 6/12: Four technology families

COMPUTING -> selected information-processing algorithms
COMMUNICATION -> quantum-state links and key distribution
SENSING / METROLOGY -> precision measurement
MATERIALS / DEVICES -> enabling hardware substrate
TRAP -> quantum technology is not computing alone

ASCII MASTER FLOW – PANEL 7/12: QKD security chain

QUANTUM STATES -> distribute key material

DISTURBANCE TEST -> reveal possible interception

AUTHENTICATED ENDPOINTS -> prevent impersonation

CLASSICAL ENCRYPTION -> protects message content

RULE -> QKD alone is not a complete secure network

ASCII MASTER FLOW – PANEL 8/12: QKD versus PQC

QKD -> quantum hardware and physical link

PQC -> classical quantum-resistant algorithms

QKD PURPOSE -> specialised key distribution

PQC PURPOSE -> broad standards and software migration

POLICY -> complementary tools with different timelines

ASCII MASTER FLOW – PANEL 9/12: NQM institution map

MINISTRY OF SCIENCE AND TECHNOLOGY -> ministerial umbrella

DST -> implementing department

NQM -> national capability mission

T-HUBS -> four vertical ecosystem anchors

RULE -> department, mission and hub are distinct

ASCII MASTER FLOW – PANEL 10/12: Four T-Hub map

IISc BENGALURU -> Quantum Computing

IIT MADRAS WITH C-DOT -> Quantum Communication

IIT BOMBAY -> Quantum Sensing and Metrology

IIT DELHI -> Quantum Materials and Devices

TRAP -> hub creation is not operational capability

MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

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ASCII MASTER FLOW – PANEL 11/12: Readiness ladder

CABINET APPROVAL -> mission authority and outlay

TARGET -> intended future capability

RESEARCH PROTOTYPE -> bounded demonstration

VALIDATED SYSTEM -> repeatable performance evidence

OPERATIONAL DEPLOYMENT -> field use with sustainment

CLOSE DISTINCTION: Qubit is not a faster classical bit, gate count is not useful...

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ASCII MASTER FLOW – PANEL 12/12: Quantum claim firewall

DEFINE -> qubit, superposition, entanglement and decoherence

CLASSIFY -> computing, communication, sensing or materials

SEPARATE -> physical and logical qubits; QKD and PQC

LABEL -> approval, target, prototype, validation and operation

VERIFY -> outlay, qubits, distance and startup claims officially

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement...

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